

Q2 (a) Fewer than four vectors will not span V .
More than four vectors will be linearly dependent.

(b) $\text{Null}(A) \perp \text{Row}(A)$, $\text{Null}(A^T) \perp \text{Col}(A)$

(c) $b \in \text{Col}(A)$

(d) (i) $u + \vec{0} = u$

$$(u_1, u_2) + (a, b) = (u_1, u_2)$$

$$(u_1 + a + 5, u_2 + b - 7) = (u_1, u_2)$$

$$u_1 + a + 5 = u_1 \Rightarrow a = -5, u_2 + b - 7 = u_2 \Rightarrow b = 7$$

$$\vec{0} = (-5, 7) \quad (2)$$

(ii) $u + u' = \vec{0}$

$$(u_1, u_2) + (a, b) = (-5, 7)$$

$$(u_1 + a + 5, u_2 + b - 7) = (-5, 7)$$

$$u_1 + a + 5 = -5 \Rightarrow a = -10 - u_1, u_2 + b - 7 = 7 \Rightarrow b = 14 - u_2$$

$$u' = (-10 - u_1, 14 - u_2) \quad (2)$$

(e)
$$\begin{bmatrix} s-3t \\ s-t \\ 2s-t \\ 4t \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ -1 \\ -1 \\ 4 \end{bmatrix} \quad (2)$$

$$u = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 0 \end{bmatrix}, v = \begin{bmatrix} -3 \\ -1 \\ -1 \\ 4 \end{bmatrix}, \dim(W) = 2 \quad (1)$$

W is a subspace of $\mathbb{R}^4$ by definition of (1)
subspace (closed under addition and scalar multiplication).

$$\frac{2}{3} + \frac{1}{3} + \frac{1}{3} = \frac{4}{3}$$

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(g)

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 9 & 3 \\ 1 & 0 & 4 \end{vmatrix} = -1 \neq 0 \quad (3)$$

$\Rightarrow S$ spans P_2 and S is linearly independent.

$\Rightarrow S$ is a basis for P_2 . (1)

(f)

$$v = (5, 2), v_1 = (1, -1), v_2 = (2, 1), v_3 = (3, 2)$$

$$k_1 v_1 + k_2 v_2 + k_3 v_3 = v$$

$$k_1(1, -1) + k_2(2, 1) + k_3(3, 2) = (5, 2) \quad (1)$$

$$\begin{bmatrix} 1 & 2 & 3 & | & 5 \\ -1 & 1 & 2 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & | & 5 \\ 0 & 3 & 5 & | & 7 \end{bmatrix} R_1 + R_2 \quad (3)$$

$$\begin{bmatrix} 1 & 2 & 3 & | & 5 \\ 0 & 1 & 5/3 & | & 7/3 \end{bmatrix} \frac{1}{3} R_2$$

$$\begin{bmatrix} 1 & 0 & -1/3 & | & 1/3 \\ 0 & 1 & 5/3 & | & 7/3 \end{bmatrix} -2R_2 + R_1$$

$$k_1 - \frac{1}{3}k_3 = \frac{1}{3} \Rightarrow k_1 = \frac{1}{3} + \frac{1}{3}k_3 \Rightarrow k_1 = \frac{1}{3} + \frac{1}{3}t$$

$$k_2 + \frac{5}{3}k_3 = \frac{7}{3} \Rightarrow k_2 = \frac{7}{3} - \frac{5}{3}k_3 \Rightarrow k_2 = \frac{7}{3} - \frac{5}{3}t$$

$$k_3 = t$$

$$t=0, k_1 = \frac{1}{3}, k_2 = \frac{7}{3}, k_3 = 0$$

$$\frac{1}{3}(1, -1) + \frac{7}{3}(2, 1) + 0(3, 2) = (5, 2)$$

$$t=1, k_1 = \frac{2}{3}, k_2 = \frac{2}{3}, k_3 = 1$$

$$\frac{2}{3}(1, -1) + \frac{2}{3}(2, 1) + 1(3, 2) = (5, 2)$$

(1)

(h)

$$[A_{3 \times 4} | I_3]$$

$$\left[\begin{array}{cccc|ccc} 1 & 5 & 2 & 4 & 1 & 0 & 0 \\ -5 & -5 & 0 & -10 & 0 & 1 & 0 \\ 7 & 1 & -3 & 11 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cccc|ccc} 1 & 5 & 2 & 4 & 1 & 0 & 0 \\ 0 & 20 & 10 & 10 & 5 & 1 & 0 \\ 0 & -34 & -17 & -17 & -7 & 0 & 1 \end{array} \right] \begin{array}{l} \\ 5R_1 + R_2 \\ -7R_1 + R_3 \end{array}$$

$$\left[\begin{array}{cccc|ccc} 1 & 5 & 2 & 4 & 1 & 0 & 0 \\ 0 & 1 & 1/2 & 1/2 & 1/4 & 1/20 & 0 \\ 0 & -34 & -17 & -17 & -7 & 0 & 1 \end{array} \right] \begin{array}{l} \\ 1/20 R_2 \\ (1) \end{array}$$

$$\left[\begin{array}{cccc|ccc} 1 & 5 & 2 & 4 & 1 & 0 & 0 \\ 0 & 1 & 1/2 & 1/2 & 1/4 & 1/20 & 0 \\ 0 & 0 & 0 & 0 & 3/2 & 34/20 & 1 \end{array} \right] 34R_2 + R_3$$

$$\left[\begin{array}{cccc|ccc} \textcircled{1} & 0 & -1/2 & 3/2 & -1/4 & -5 & 0 \\ 0 & \textcircled{1} & 1/2 & 1/2 & 1/4 & 1/20 & 0 \\ 0 & 0 & 0 & 0 & 3/2 & 34/20 & 1 \end{array} \right] \begin{array}{l} -5R_2 + R_1 \\ \\ \end{array}$$

Null(A)

$$x_1 - 1/2 x_3 + 3/2 x_4 = 0 \Rightarrow x_1 = 1/2 x_3 - 3/2 x_4$$

$$x_2 + 1/2 x_3 + 1/2 x_4 = 0 \Rightarrow x_2 = -1/2 x_3 - 1/2 x_4$$

$$x_1 = 1/2 s - 3/2 t$$

$$x_2 = -1/2 s - 1/2 t$$

$$x_3 = s, x_4 = t$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1/2 s - 3/2 t \\ -1/2 s - 1/2 t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3/2 \\ -1/2 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Basis for Null(A)} = \left\{ \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3/2 \\ -1/2 \\ 0 \\ 1 \end{bmatrix} \right\} \quad (1)$$

$$\text{For Col}(A) = \left\{ \begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix}, \begin{bmatrix} 5 \\ -5 \\ 1 \end{bmatrix} \right\} \quad (1)$$

$$\text{For Row}(A) = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1/2 \\ 3/2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1/2 \\ 1/2 \end{bmatrix} \right\} \text{ or } \left\{ \begin{bmatrix} 1 \\ 5 \\ 2 \\ 4 \end{bmatrix}, \begin{bmatrix} -5 \\ -5 \\ 0 \\ -10 \end{bmatrix} \right\} \quad (1)$$

$$\text{For Null}(A^T) = \left\{ \begin{bmatrix} 3/2 \\ 3/2 \\ 1 \end{bmatrix} \right\} \quad (1) \quad 2(i) \quad P = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad (1)$$

$$B = \text{RGB basis} = \{e_1, e_2, e_3\}, \quad B' = \{b_1, b_2, b_3\}$$

$$(ii) \quad P_{B \rightarrow B'} = \begin{bmatrix} 1 & 0 & 1 & | & 1 & 0 & 0 \\ 1 & 1 & 0 & | & 0 & 1 & 0 \\ 0 & 1 & 1 & | & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} 1 & 0 & 1 & | & 1 & 0 & 0 \\ 0 & 1 & -1 & | & -1 & 1 & 0 \\ 0 & 1 & 1 & | & 0 & 0 & 1 \end{bmatrix} \xrightarrow{-R_1+R_2} \begin{bmatrix} 1 & 0 & 1 & | & 1 & 0 & 0 \\ 0 & 1 & -1 & | & -1 & 1 & 0 \\ 0 & 0 & 2 & | & 1 & -1 & 1 \end{bmatrix} \xrightarrow{-R_2+R_3}$$

$$\begin{bmatrix} 1 & 0 & 1 & | & 1 & 0 & 0 \\ 0 & 1 & -1 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & 1/2 & -1/2 & 1/2 \end{bmatrix} \xrightarrow{-R_3+R_1} \begin{bmatrix} 1 & 0 & 0 & | & 1/2 & 1/2 & -1/2 \\ 0 & 1 & -1 & | & -1 & 1 & 0 \\ 0 & 0 & 1 & | & 1/2 & -1/2 & 1/2 \end{bmatrix} \xrightarrow{R_3+R_2}$$

$$P_{B \rightarrow B'} = \begin{bmatrix} 1/2 & 1/2 & -1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & 1/2 \end{bmatrix} \quad (2)$$

$$(iii) \quad [V]_{B'} = P_{B \rightarrow B'} [V]_{RGB} = \begin{bmatrix} 1/2 & 1/2 & -1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 0.3 \\ 0.6 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.35 \\ 0.25 \\ -0.05 \end{bmatrix} \quad (1)$$